ABERDARE TECHNICAL COLLEGE

COMPUTER REPAIR AND MAINTENANCE

PRACTICAL ASSESSMENT OBSERVATION CHECKLIST 2

Unit of Competency: Perform Computer Repair and Maintenance

Unit Code: 0714 351 04A

Element 3: Test Computer Component Functionality

Candidate Name: ____________________________

Reg. No: ____________________________

Date: ____________________________

Assessor: ____________________________

Duration: 2 Hours

CUTTING LIST (MATERIALS / EQUIPMENT USED)

No.	Item	Quantity	Provided By
1	Desktop Computer (Faulty Unit)	1	College
2	Monitor	1	College
3	Keyboard	1	College
4	Mouse	1	College
5	Diagnostic Software (Memtest86, CrystalDiskInfo)	1 set	Internet

6	Network Cable	1	College
7	Screwdriver Set	1 set	College
8	Replacement Components (RAM/HDD/SSD etc.)	As required	College
9	Power Extension Cable	1	College
10	USB Flash Drive	1	Student/College

PRACTICAL OBSERVATION CHECKLIST

TASK 1: PERFORM POWER-ON SELF-TEST (POST) – 10 MARKS

No.	Item to be evaluated	Marks	Score	Remarks
1	Observes safety precautions	1		
2	Powers on computer correctly	2		
3	Observes POST messages/beep codes	2		
4	Identifies detected hardware	2		
5	Determines POST success/failure	2		
6	Records findings	1		
	Subtotal	**10**		

TASK 2: COMPONENT TESTING – 20 MARKS

No.	Item to be evaluated	Marks	Score	Remarks
1	RAM test using MemTest86	2		
2	Records RAM results	2		
3	HDD/SSD test using CrystalDiskInfo	2		
4	Records storage results	2		
5	Keyboard testing	2		
6	Records keyboard results	2		
7	Mouse testing	2		
8	Records mouse results	2		
9	Monitor testing	2		
10	Records monitor results	2		
	Subtotal	**20**		

TASK 3: REPAIRED/REPLACED COMPONENT TEST – 10 MARKS

No.	Activity	Marks	Score	Remarks
1	Identifies replaced component	2		

2	Correct installation/reconnection	2		
3	Performs functionality test	2		
4	Confirms fault correction	2		
5	Records results	2		
	Subtotal	**10**		

TASK 4: EVALUATION – 5 MARKS

No.	Activity	Marks	Score	Remarks
1	Analyzes results	1		
2	Identifies faults	1		
3	Determines causes	1		
4	Suggests corrective actions	1		
5	Determines system fitness	1		
	Subtotal	**5**		

TASK 5: FUNCTIONALITY REPORT – 5 MARKS

No.	Activity	Marks	Score	Remarks
1	Candidate details included	1		

2	Problem and POST findings included	1		
3	Components and tools listed	1		
4	Results and evaluation included	1		
5	Recommendations and conclusion	1		
	Subtotal	**5**		

SUMMARY OF MARKS (50 MARKS)

Task	Marks	Score
Task 1	10	
Task 2	20	
Task 3	10	
Task 4	5	
Task 5	5	
TOTAL	**50**	

PRODUCT CHECKLIST (10)

No.	Item to be evaluated	Marks	Score	Remarks
1	POST findings recorded correctly	2		
2	RAM test results documented	1		
3	HDD/SSD test results documented	1		
4	Keyboard, mouse, monitor results documented	2		
5	Repaired/replaced component results recorded	1		
6	Evaluation and recommendations included	1		
7	Professional report formatting and completeness	2		
	TOTAL	**10**		

ORAL QUESTIONING CHECKLIST (30 MARKS)

No.	Oral Question	Expected Response	Marks	Score	Remarks
1	What is POST in a computer system?	Power-On Self-Test performed during startup.	2		
2	Why is POST important?	Detects hardware faults before loading the operating system.	2		
3	What may cause a computer to fail POST?	Faulty RAM, motherboard, CPU, PSU, etc.	2		
4	What do beep codes during POST indicate?	Hardware errors detected by BIOS/UEFI.	2		
5	What is the purpose of testing computer components after repair?	To verify correct operation and reliability.	2		

6	State two importance of testing repaired or replaced components.	Confirms functionality and ensures customer satisfaction.	2		
7	What is a diagnostic test?	A test used to identify hardware faults.	2		
8	Name two computer components commonly tested after repair.	RAM, hard drive/SSD, motherboard, PSU, CPU, etc.	2		
9	How would you test the functionality of RAM?	Using memory diagnostic tools or swapping with known good RAM.	2		
10	How would you test a hard disk drive after replacement?	Run disk diagnostics, SMART tests, and file operations.	2		
11	What is meant by a known-good component testing technique?	Replacing suspected component with a working one for comparison.	2		
12	Why should peripheral devices be tested after repair?	To ensure they communicate and function correctly.	2		
13	What information should be recorded during component testing?	Component tested, test results, observations, and actions taken.	2		
14	What is meant by evaluation of test results?	Analyzing results to determine whether components pass or fail.	2		
15	Why is a functionality report important after testing?	Provides evidence of repair, testing, and component status.	2		

Total Marks: 30

COMPETENCY DECISION

☐ **COMPETENT (C)** – Candidate has met all required standards (≥ 50 marks + satisfactory product evidence)

☐ **NOT YET COMPETENT (NYC)** – Candidate has not met required standards and requires reassessment

ASSESSOR COMMENTS

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Assessor Name: ______________________

Signature: ___________________________

Date: _______________________________